# Danni Liu

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## Experience

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| 11/2025 – now | **Postdoctoral Researcher**, *Karlsruhe Institute of Technology (KIT)*, Karlsruhe, Germany |
| 02/2020 – 11/2025 | **Doctoral Researcher**, *KIT*, Karlsruhe, Germany<br>Until 06/2022 at *Maastricht University*, Netherlands; moved to KIT with advisor<br>Ph.D. in Computer Science (summa cum laude), advised by Jan Niehues<br>*Thesis:* Generalizable Representations for Cross-Lingual Transfer Learning |
| 07/2021 – 01/2022 | **Research Intern**, *Facebook / Meta AI*, Remote<br>10/2021 – 01/2022: employed as contingent worker via Blue Lynx<br>*Project:* Low-latency TTS for speech-to-speech translation (Interspeech 2022 [6]) |
| 03/2019 – 08/2019 | **Research Intern**, *Nuance Communications (now Microsoft)*, Aachen, Germany |
| 08/2015 – 01/2019 | **Trainee** → **Business Intelligence Consultant**, *Open Line*, Maastricht, Netherlands |

## Education

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| 02/2020 – 11/2025 | **Ph.D. Computer Science** (summa cum laude)<br>*Karlsruhe Institute of Technology*, Germany |
| 02/2018 – 01/2020 | **M.Sc. Artificial Intelligence** (cum laude)<br>*Maastricht University*, Netherlands<br>*Thesis:* Low-Latency End-to-End Speech Recognition with Enhanced Readability |
| 09/2014 – 08/2017 | **B.Sc. Data Science and Knowledge Engineering** (cum laude)<br>*Maastricht University*, Netherlands |
| 09/2011 – 08/2014 | **B.Sc. International Business Administration** (cum laude)<br>*Erasmus University Rotterdam*, Netherlands |

## Talks

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| 09/2025 | **AiMat Summer School**, "A Mini Introduction to Large Language Models" [slides] |
| | **University of Toronto & Ontario Tech University**, two talks: |
| 09/2024 | "Retrieval-Augmented Approaches for Low-Resource Conditions in Speech Translation [slides] |
| 07/2024 | "Learning from Less: Machine Translation and Summarization in Low-Resource Settings [slides] |
| 12/2022 | **IWSLT Lecture Series**, "Learning Shared Representations for Translation in Low-Resource Conditions" [video] [slides] |
| 02/2022 | **Meta AI**, "Studying Shared Representations in NMT through Zero-Shot Translation" [slides] |

## Mentoring & Teaching

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| 2020 – now | **Supervision of Bachelor/Master's Theses & Internships**<br>Leading to 8 publications, incl. EMNLP and NAACL [21, 16, 15, 13, 11, 9, 5, 4] |
| 2022 – now | **Tutor**, *Karlsruhe Institute of Technology*, Germany<br>*Courses:* Natural Language Processing, Machine Translation (guest lecturer) |
| 2019 – 2021 | **Teaching Assistant**, *Maastricht University*, Netherlands<br>*Courses:* Advanced Natural Language Processing, Probability and Statistics |

## Publications

23. Johannes Wesch, **Danni Liu**, and Jan Niehues. 2026. KV$^2$: A Self-Refining KV Cache. *Under Review*

22. Maike Züfle, **Danni Liu**, Vilém Zouhar, and Jan Niehues. 2026. Why We Need Speech to Evaluate Speech Translation. *EMNLP*

21. Leonard Popp, **Danni Liu**, and Jan Niehues. 2026. Predicting Steering Vectors and Adapter Weights for Few-Shot Author-Style Transfer. *W-NUT Workshop @ EMNLP*

20. Andreas Hochlehnert, **Danni Liu**, Jan Niehues, Shankar Kumar, and Felix Stahlberg. 2026. On the Importance of Trivial Baselines: Re-evaluating LoRA Adapter Transfer for Generative Tasks. *CATS Workshop @ ICML*

19. Sara Papi, Maike Züfle, Marco Gaido, Beatrice Savoldi, **Danni Liu**, Ioannis Douros, Luisa Bentivogli, and Jan Niehues. 2026. MCIF: Multimodal Crosslingual Instruction-Following Benchmark from Scientific Talks. *ICLR*

18. **Danni Liu** and Jan Niehues. 2025. Conditions for Catastrophic Forgetting in Multilingual Translation. [poster] *MRL Workshop @ EMNLP*

17. **Danni Liu** and Jan Niehues. 2025. Middle-Layer Representation Alignment for Cross-Lingual Transfer in Fine-Tuned LLMs. [poster] *ACL*

16. Hyunji Lee, **Danni Liu**, Supriti Sinhamahapatra, and Jan Niehues. 2025. How do Multimodal Foundation Models Encode Text and Speech? An Analysis of Cross-Lingual and Cross-Modal Representations. [poster] *NAACL*

15. Siqi Li*, **Danni Liu***, and Jan Niehues. 2024. Optimizing Rare Word Accuracy in Direct Speech Translation with a Retrieval-and-Demonstration Approach. (* indicates equal contribution) *EMNLP*

14. Tu Anh Dinh, Carlos Mullov, Leonard Bärmann, Zhaolin Li, **Danni Liu**, ..., and Jan Niehues. 2024. SciEx: Benchmarking Large Language Models on Scientific Exams with Human Expert Grading and Automatic Grading. *EMNLP*

13. Lukas Hilgert, **Danni Liu**, and Jan Niehues. 2024. Evaluating and Training Long-Context Large Language Models for Question Answering on Scientific Papers. *CustomNLP4U Workshop @ EMNLP*

12. **Danni Liu** and Jan Niehues. 2024. Recent Highlights in Multilingual and Multimodal Speech Translation. [slides] *IWSLT*

11. Vladimir Solovyev, **Danni Liu**, and Jan Niehues. 2024. Language-Independent Representations Improve Zero-Shot Summarization. *NAACL*

10. **Danni Liu** and Jan Niehues. 2024. How Transferable are Attribute Controllers on Pretrained Multilingual Translation Models?. [slides] [poster] *EACL*

9. Yunus Demirag, **Danni Liu**, and Jan Niehues. 2024. Benchmarking Diffusion Models for Machine Translation. *Student Research Workshop @ EACL*

8. Christian Huber, Tu Anh Dinh, Carlos Mullov, Ngoc-Quan Pham, Thai Binh Nguyen, Fabian Retkowski, Stefan Constantin, Enes Ugan, **Danni Liu**, Zhaolin Li, Sai Koneru, Jan Niehues, and Alexander Waibel. 2023. End-to-End Evaluation for Low-Latency Simultaneous Speech Translation. *EMNLP Demo*

7. **Danni Liu** and Jan Niehues. 2022. Learning an Artificial Language for Knowledge-Sharing in Multilingual Translation. [video] [slides] *WMT*

6. **Danni Liu**, Changhan Wang, Hongyu Gong, Xutai Ma, Yun Tang, and Juan Pino. 2022. From Start to Finish: Latency Reduction Strategies for Incremental Speech Synthesis in Simultaneous Speech-to-Speech Translation. [poster] *Interspeech*

5. Tu Anh Dinh, **Danni Liu**, and Jan Niehues. 2022. Tackling Data Scarcity in Speech Translation using Zero-shot Multilingual Machine Translation Techniques. *ICASSP*

4. Sai Koneru, **Danni Liu**, and Jan Niehues. 2021. Unsupervised Machine Translation On Dravidian Languages. *DravidianLangTech Workshop @ EACL*

3. **Danni Liu**, Jan Niehues, James Cross, Francisco Guzmán, and Xian Li. 2021. Improving Zero-shot Translation by Disentangling Positional Information. [video] [slides] *ACL*

2. **Danni Liu**, Gerasimos Spanakis, and Jan Niehues. 2020. Low-Latency Sequence-to-Sequence Speech Recognition and Translation by Partial Hypothesis Selection. [video] [slides] *Interspeech*

1. **Danni Liu**, Jan Niehues, and Gerasimos Spanakis. 2020. Adapting End-to-End Speech Recognition for Readable Subtitles. [video] [slides] *IWSLT*

### Shared Tasks (Selected)

S7. **Danni Liu**, Thai Binh Nguyen, Sai Koneru, Enes Yavuz Ugan, Ngoc-Quan Pham, Tuan Nam Nguyen, Tu Anh Dinh, Carlos Mullov, Alexander Waibel, and Jan Niehues. 2023. KIT's Multilingual Speech Translation System for IWSLT 2023. [poster] 🏆 **Winning system in multilingual track (constrained)** *IWSLT*

S6. Peter Polák, **Danni Liu**, Ngoc-Quan Pham, Jan Niehues, Alexander Waibel, and Ondřej Bojar. 2023. Towards Efficient Simultaneous Speech Translation: CUNI-KIT System for Simultaneous Track at IWSLT 2023. *IWSLT*

S5. Abteen Ebrahimi, ..., **Danni Liu**, et al.. 2022. Findings of the Second AmericasNLP Competition on Speech-to-Text Translation. 🏆 **Winning system in speech translation and machine translation tracks** *NeurIPS 2022 Competition Track*

S4. Ngoc-Quan Pham, Tuan Nam Nguyen, Thai-Binh Nguyen, **Danni Liu**, Carlos Mullov, Jan Niehues, and Alexander Waibel. 2022. Effective combination of pretrained models - KIT@IWSLT 2022 *IWSLT*

S3. Peter Polák, Ngoc-Quan Pham, Tuan Nam Nguyen, **Danni Liu**, Carlos Mullov, Jan Niehues, Ondřej Bojar, and Alexander Waibel. 2022. CUNI-KIT System for Simultaneous Speech Translation Task at IWSLT 2022. 🏆 **Winning system in simultaneous translation track** *IWSLT*

S2. **Danni Liu** and Jan Niehues. 2021. Maastricht University's Large-Scale Multilingual Machine Translation System for WMT 2021. [poster] 🏛 **Best performing system from university participants** *WMT*

S1. **Danni Liu** and Jan Niehues. 2021. Maastricht University's Multilingual Speech Translation System for IWSLT 2021. [video] [slides] *IWSLT*

## Service

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| Awards | Best Reviewer Award, Interspeech 2026; 6× Great Reviewer, ACL Rolling Review |
| Reviewer | ACL Rolling Review (2023–now), NeurIPS (2026), ICASSP (2024–now), Interspeech (2024–now), WMT (2026, 2025, 2022), various workshops |

— Last updated on 30 September, 2026 —